1

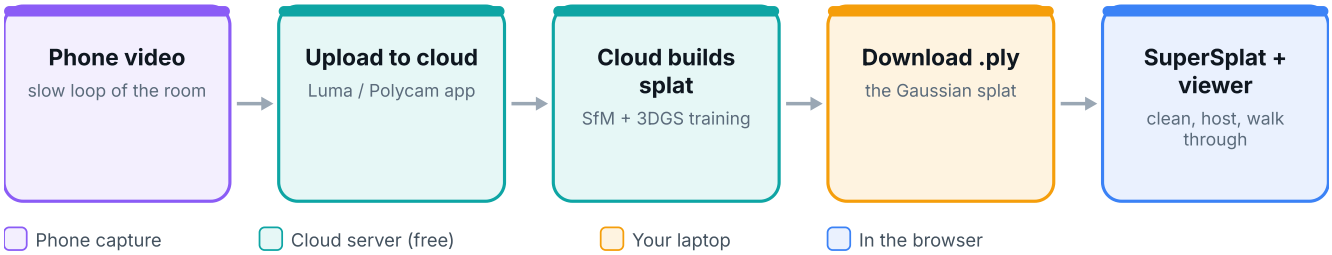
OPTION 1 · RECONSTRUCTION

Cloud Gaussian Splatting

Luma AI / Polycam — the cloud does the hard 3D work.

★ Recommended

In one line: record a video, let a **free cloud service** turn it into a photorealistic Gaussian splat, then view & publish it in the browser. You do almost no computing — their servers do the hard part.



The heavy 3D training happens on the cloud server, not your phone or laptop.

How it works, step by step

1. **Capture** a 30–60s video of the room with the **Luma AI** or **Polycam** app (or record normally and upload). Move slowly, keep lots of overlap.
2. The app **uploads** the frames to the service's cloud.
3. Their **cloud GPUs** estimate camera positions and **train the Gaussian splat** (a few minutes) — the step that keeps failing on-phone is done on strong hardware here.
4. You **download the result as a `.ply`** splat (Luma also gives its own web link).
5. I load it into **SuperSplat** to crop junk + compress, then drop it into **our web viewer** on GitHub Pages — that link is the Round-1 deliverable.

Trade-offs

✓ Strengths

- Photorealistic — true 3D Gaussian splat
- Almost no work/compute on your side
- Cloud training rarely fails (vs on-phone)
- Works from any ordinary video
- Free tiers are enough for one room

✗ Weaknesses

- Needs internet + an account
- Free tier has limits (scans/month, resolution)
- You wait a few minutes for cloud processing
- Your video leaves your device (privacy)

Tools: Luma AI, Polycam | Input: phone video | Output: .ply splat | Your compute: ~none | Cost: free tier

WHAT WE ACTUALLY DID

Not attempted yet — this is the **recommended next step** after the on-device Scaniverse route failed to train. It sidesteps exactly the failure we hit.

VERDICT FOR THE COMPETITION

Best choice for Round 1. Highest quality for the least effort and the most reliable. This should be plan A.